

Homework 16

Please read Section 5.2 of Durrett. Skip Section 5.2.1 and continue with Section 5.2.2 to learn more about medians.

Example 5.10 in Section 5.2 is nice. Durrett calculates the distribution function for a distribution that is not continuous. The distribution function is not continuous, since it has jumps at 1, 2, and 3. Continuous random variables, on the other hand, have continuous distribution functions. Textbooks that focus more on theory sometimes **define** continuous random variables to be random variables with continuous distribution functions.

Please work the following exercises from Chapter 5:

2

Let f be the function defined by $f(x) = 0$ if $x \leq -1$, $f(x) = c(1 - x^2)$ for a constant c , and $f(x) = 0$ if $x \geq 1$. To make the function a density function, we need $\int_{-\infty}^{\infty} f(x)dx = 1$. Because f is the zero function outside of the interval $(-1, 1)$, this integral is $\int_{-1}^1 f(x)dx = \int_{-1}^1 c(1 - x^2)dx$. We thus have

$$\begin{aligned}\int_{-1}^1 c(1 - x^2)dx &= 1 \\ \int_{-1}^1 (c - cx^2)dx &= 1 \\ \left[cx - \frac{1}{3}cx^3 \right]_{x=-1}^{x=1} &= 1 \\ c - \frac{1}{3}c - \left((-c) + \frac{1}{3}c \right) &= 1 \\ 2c - \frac{2}{3}c &= 1 \\ \frac{4}{3}c &= 1 \\ c &= \frac{3}{4}.\end{aligned}$$

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The random variable X has the density function f defined by $f(x) = 0$ if $x \leq 0$, $f(x) = \frac{x^2}{9}$ if $0 < x < 3$, and $f(x) = 0$ if $x \geq 3$. We calculate $E[X]$, $E[X^2]$, and $\text{Var}(X)$ as follows:

$$\begin{aligned}E[X] &= \int_0^3 xf(x)dx \\ &= \int_0^3 x \left(\frac{x^2}{9} \right) dx \\ &= \int_0^3 \frac{1}{9}x^3 dx\end{aligned}$$

$$\begin{aligned}
&= \left[\frac{1}{36} x^4 \right]_{x=0}^{x=3} \\
&= \frac{81}{36} = \frac{9}{4} = 2.25.
\end{aligned}$$

$$\begin{aligned}
E[X^2] &= \int_0^3 x^2 f(x) dx \\
&= \int_0^3 x^2 \left(\frac{x^2}{9} \right) dx \\
&= \int_0^3 \frac{1}{9} x^4 dx \\
&= \left[\frac{1}{45} x^5 \right]_{x=0}^{x=3} \\
&= \frac{243}{45} = \frac{27}{5} = 5.4
\end{aligned}$$

$$\text{Var}(X) = E[X^2] - E[X]^2 = \frac{27}{5} - \left(\frac{9}{4} \right)^2 = \frac{27}{5} - \frac{81}{16} = \frac{432 - 405}{80} = \frac{27}{80} = 0.3375.$$

6 (As Durrett mentions in Section 5.2, the density function for a continuous distribution is the derivative of the distribution function. Use that fact to write a formula for the density function. In your formula for f , you will need three cases, including values for $f(x)$ when $x < 0$ and when $x > 1$.)

Let F be defined by $F(x) = 0$ if $x \leq 0$, $F(x) = 3x^2 - 2x^3$ if $0 < x < 1$, and $F(x) = 1$ if $x \geq 1$. Durrett tells us that F is a distribution function. The associated density function is the derivative F' . This function is defined by $F'(x) = 0$ if $x \leq 0$, $F'(x) = 6x - 6x^2$ if $0 < x < 1$, and $F'(x) = 0$ if $x \geq 1$.

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We are given the density function f defined by $f(x) = 0$ if $x \leq 0$, $f(x) = 4x^3$ if $0 < x < 1$, and $f(x) = 0$ if $x \geq 1$.

part a. The distribution function F is defined by $F(x) = \int_{-\infty}^x f(t) dt$. For $x \leq 0$, we have $F(x) = 0$. If $0 < x < 1$, we have

$$\begin{aligned}
F(x) &= \int_{-\infty}^x f(t) dt = \int_0^x f(t) dt \\
&= \int_0^x 4t^3 dt \\
&= [t^4]_{t=0}^{t=x} \\
&= x^4.
\end{aligned}$$

And if $x \geq 1$, we have $F(x) = 1$.

part b. We can calculate $P\left(X < \frac{1}{2}\right)$ as $F\left(\frac{1}{2}\right)$, since X is a continuous random variable ($P\left(X < \frac{1}{2}\right) = P\left(X \leq \frac{1}{2}\right)$). Therefore, $P\left(X < \frac{1}{2}\right) = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$.

part c. We can calculate $P\left(\frac{1}{3} < x < \frac{2}{3}\right)$ as $F\left(\frac{2}{3}\right) - F\left(\frac{1}{3}\right) = \left(\frac{2}{3}\right)^4 - \left(\frac{1}{3}\right)^4 = \frac{16}{81} - \frac{1}{81} = \frac{15}{81} = \frac{5}{27}$. We could also calculate this probability as $\int_{\frac{1}{3}}^{\frac{2}{3}} f(x)dx$. Using x^4 as an antiderivative of $f(x)$ on the interval $\left(\frac{1}{3}, \frac{2}{3}\right)$ gives the same value for the integral (and thus for the probability) as we found using the distribution function.

part d. To find the median of X , we solve $F(m) = \frac{1}{2}$ for m . Any solution m of that equation must be between 0 and 1. Using the formula for F in this range of inputs, we find $m^4 = \frac{1}{2}$ and thus $m = \left(\frac{1}{2}\right)^{\frac{1}{4}}$.

Please also answer the following questions:

1. Hood College is next to Frederick Health Hospital (FHH), which is the central medical facility in Frederick County. According to publicity material from FHH, there are about 2400 babies born each year at the hospital.
 - a. Suppose you want to model the number of babies born each day in the hospital with a random variable N . What distribution would you propose to use for N ? This is a review question! To answer it, think about discrete distributions that we used to model count data. Pick the parameter or parameters for your distribution to match the average number of babies born each day (2400/365).

$N \sim \text{Poisson}(2400/365)$

- b. According to your model, what is the standard deviation (square root of variance) in the number of babies born each day at the hospital? This is also a review question. You will use a formula for the variance of the distribution that you picked in the previous question.

$$\sigma(N) = \sqrt{\frac{2400}{365}} \approx 2.56 \text{ babies}$$

2. Suppose we want to model the wait time between arrivals of babies in the hospital nursery.
 - a. According to the data that FHH has provided (2400 babies born per year), what is the average gap in hours between births of babies at this hospital?

The average gap is the number of hours per year divided by the number of babies born per year. That average gap is $(365)(24)/2400 = 73/20 = 3.65$ hours (per baby).

- b. Let X be a random variable representing the wait time between babies, measured in hours. We will model the wait time by giving X the exponential(λ) distribution for a suitable λ . According to the formulas from Durrett Section 5.1.1, what is the expected value of X , if $X \sim \text{exponential}(\lambda)$?

$$E[X] = 1/\lambda$$

- c. Choose the parameter λ to make $E[X]$ match the average gap in hours between births at FHH. What is your value of λ ?

We solve $\frac{1}{\lambda} = \frac{73}{20}$ to get $\lambda = \frac{20}{73} \approx 0.274$ babies per hour.

- d. According to your model, what is the median wait time between babies? Durrett gave a formula for this median (for a continuous random variable with the exponential(λ) distribution) in the assigned reading about medians!

Durrett provides a formula for the median of an exponentially distributed random variable: if $X \sim \text{exponential}(\lambda)$, then the median of X is $\ln(2) / \lambda$. That works out to be $\ln(2) \left(\frac{73}{20} \right) \approx 2.53$ hours.